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When the atmosphere is unstable, parcels of air rise upward, which can lead to the formation of thunderstorms. Atmospheric soundings are used to calculate the convective available potential energy (CAPE), which equals the work done by the buoyant force W_B divided by the mass m of the air parcel.

$$CAPE = \frac{W_B}{m}$$

Equation 1

The data collected from weather balloons can also be used to create a diagram called a hodograph, which represents wind velocity vectors at different heights in the atmosphere.

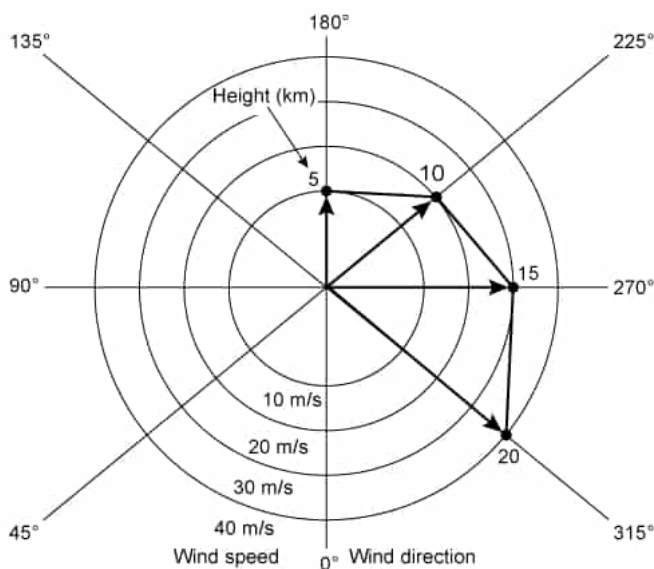


Figure 1 An example hodograph

Concentric circles of increasing radius about the origin show the wind speed, and the horizontal wind direction is represented by the angle relative to a reference direction denoted as 0°. The hodograph in Figure 1 shows the wind speeds at 5 km height intervals above the Earth's surface. The hodograph line connects the tips of each wind vector at different heights.

Hodographs indicate wind shear in the atmosphere, which is an important factor in tornado formation.

A spherical weather balloon is filled with helium gas with a density of about 0.18 g/L. If the diameter of the balloon is 20 m, what is the approximate mass of the helium in the balloon? (Note: The volume of a sphere is $\frac{4}{3}\pi r^3$ and 1 L = 1,000 cm³.)

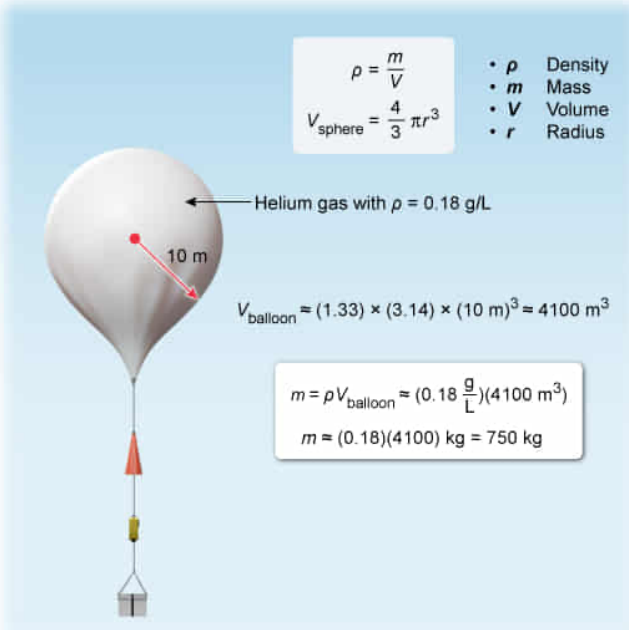
- ☐ A. 750,000 kg [11%]
☐ B. 6,000 kg [14%]
✓ ☒ C. 750 kg [45%]
☐ D. 0.75 kg [28%]

Correct

44% answered correctly

Explanation:

Calculating the mass of helium gas in a weather balloon using density and volume



Helium gas with $\rho = 0.18 \text{ g/L}$

10 m

$V_{\text{sphere}} = \frac{4}{3}\pi r^3$

$V_{\text{balloon}} \approx (1.33) \times (3.14) \times (10 \text{ m})^3 \approx 4100 \text{ m}^3$

$m = \rho V_{\text{balloon}} \approx (0.18 \frac{\text{g}}{\text{L}})(4100 \text{ m}^3)$
 $m = (0.18)(4100) \text{ kg} = 750 \text{ kg}$

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The **density** ρ of an object is defined as the ratio of the object's **mass** m and **volume** V :

$$\rho = \frac{m}{V}$$

If the density and volume of the object are known, then the mass of the substance can be determined by calculating the product of ρ and V :

$$m = \rho V$$

In this question, helium gas with a density of about 0.18 g/L fills a spherical balloon of diameter 20 m, or, equivalently, of radius 10 m. The volume of a spherical region of radius r equals:

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3$$

Inserting the value of r into this formula yields:

$$V_{\text{balloon}} \approx (1.33) \times (3.14) \times (10 \text{ m})^3$$

$$V_{\text{balloon}} \approx (1.33) \times (3.14) \times 1,000 \text{ m}^3$$

$$V_{\text{balloon}} \approx 4100 \text{ m}^3$$

Hence, the formula for m implies that:

$$m \approx \left(0.18 \frac{\text{g}}{\text{L}}\right)(4,100 \text{ m}^3)$$

Using the unit conversion between liters and cubic centimeters, 1 L = 1,000 cm³:

$$m = \left(\frac{0.18 \text{ g}}{1,000 \text{ cm}^3}\right)(4,100 \text{ m}^3)$$

Furthermore, because 1 m = 100 cm:

$$1 \text{ m}^3 = (100)^3 \text{ cm}^3 = 1 \times 10^6 \text{ cm}^3$$

Therefore:

$$m \approx \left(\frac{0.18 \text{ g}}{1,000 \text{ cm}^3}\right)(4,100 \times 10^6 \text{ cm}^3)$$

$$m \approx (0.18)(4,100 \times 10^3) \text{ g}$$

Finally, the conversion between grams and kilograms (1,000 g = 1 kg) yields:

$$m \approx (0.18)(4,100) \text{ kg} \approx 750 \text{ kg}$$

(Choice A) 750,000 kg results from neglecting to divide by 1000 in the conversion from g to kg.

(Choice B) 6,000 kg follows from incorrectly using 20 m (the balloon diameter) as the radius of the sphere.

(Choice D) 0.75 kg results from diving by an extra factor of 1000 in the conversion between units.

Educational objective:

Density is the mass per unit volume of a substance. The total mass of an object can be determined from the product of the object's density and volume.

Time spent:0 sec

Subject:
Physics

QID:404110

System:
Fluids

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Equation 1

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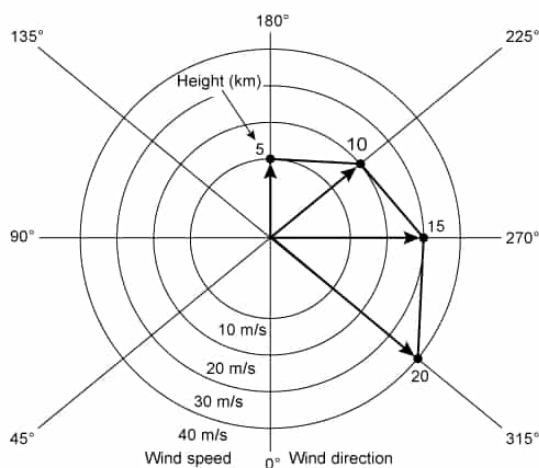


Figure 1 An example hodograph

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Balloons can be used to carry loads upward in the air using buoyant force. A balloon of volume $V = 14,000 \text{ m}^3$ is filled with hydrogen gas of density $\rho_h = 0.08376 \text{ kg/m}^3$. The density of air is $\rho_{\text{air}} = 1.204 \text{ kg/m}^3$. Which expression gives the maximum possible mass of the load that can be lifted by the balloon?

- ☐ A. $(0.08376) \times (9.80) \times (14,000)$ [8%]
- ☐ B. $(1.204 - 0.08376) \times (9.80) \times (14,000)$ [42%]

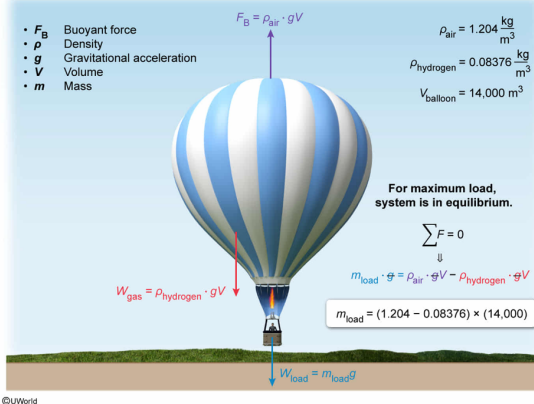
- ✓ ☒ C. $(1.204 - 0.08376) \times (14,000)$ [42%]
☐ D. $(1.204 + 0.08376) \times (14,000)$ [6%]

Correct

42% answered correctly

Explanation:

Determining the maximum mass that can be lifted by a hydrogen balloon



Archimedes' principle states that an object submersed in a fluid experiences an upward **buoyant force** F_B equal to the **weight** of the fluid displaced by the object. Consequently, F_B equals the product of the fluid density ρ , the gravitational acceleration g , and the amount of volume V displaced:

$$F_B = \rho \cdot gV$$

In this question, there are three forces acting on the balloon system: the F_B of the air, the weight W_{gas} of the hydrogen gas in the balloon, and the weight of the load W_{load} . The upward-directed F_B equals the product of the density of air ρ and the V of the balloon:

$$F_B = \rho_{\text{air}} \cdot gV_{\text{balloon}}$$

Furthermore, the weight of the hydrogen in the balloon is the product of the density of the hydrogen gas ρ and the V of the balloon:

$$W_{\text{gas}} = \rho_{\text{hydrogen}} \cdot gV_{\text{balloon}}$$

According to **Newton's second law of motion**, the sum of all the forces on the balloon is proportional to the balloon's upward acceleration a :

$$\sum F \propto a$$

$$F_B - W_{\text{gas}} - W_{\text{load}} \propto a$$

$$\rho_{\text{air}} \cdot gV_{\text{balloon}} - \rho_{\text{hydrogen}} \cdot gV_{\text{balloon}} - W_{\text{load}} \propto a$$

As the weight of the load is increased, a decreases to zero, which is the weight at which the balloon will not rise upward from the ground and will remain in equilibrium. Solving the above equation for the maximum W_{load} gives:

$$W_{\text{load}} = \rho_{\text{air}} \cdot g V_{\text{balloon}} - \rho_{\text{hydrogen}} \cdot g V_{\text{balloon}}$$

Because $W_{\text{load}} = m_{\text{load}} g$, it follows that:

$$m_{\text{load}} \cdot g = \rho_{\text{air}} \cdot g V_{\text{balloon}} - \rho_{\text{hydrogen}} \cdot g V_{\text{balloon}}$$

Therefore, inserting the values of the densities and the volume of the balloon yields:

$$m_{\text{load}} = (1.204 - 0.08376) \times (14,000)$$

(Choice A) $(0.08376) \times (9.80) \times (14,000)$ is the total weight of the hydrogen gas in the balloon.

(Choice B) $(1.204 - 0.08376) \times (9.80) \times (14,000)$ includes an extra factor of $g = 9.80 \text{ m/s}^2$ and is the maximum weight of the load, not its mass.

(Choice D) $(1.204 + 0.08376) \times (14,000)$ incorrectly sums the densities of the air and the hydrogen.

Educational objective:

An object submersed in a fluid experiences an upward buoyant force equal to the weight of the fluid displaced by the object. An object is in equilibrium if the sum of the forces exerted on the object equals zero.

Time spent: 0 sec

Subject:
Physics

QID: 404111

System:
Fluids

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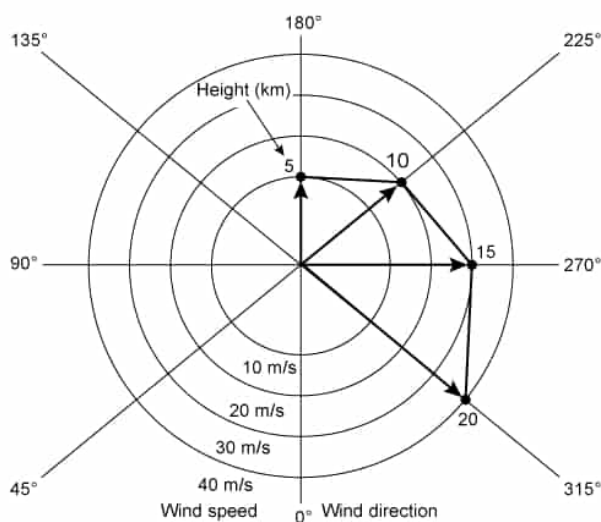


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In a hot air balloon, a flame heats the air inside the balloon and the balloon rises

upward. Why does heating the air cause the balloon to rise?

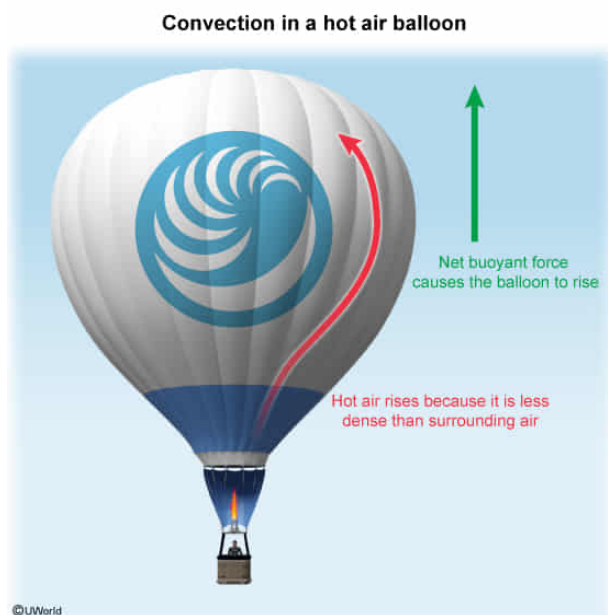


- ☒ A. The heated air becomes less dense, and convection causes it to rise. [87%]
- ☐ B. The heated air becomes denser, and convection causes it to rise. [5%]
- ☐ C. The radiation created by the flame pushes the air upward. [1%]
- ☐ D. The heat transferred by conduction pushes the air upward. [5%]

Correct

88% answered correctly

Explanation:



When two regions have different temperatures, there is a spontaneous transfer of energy, known as **heat**, between the regions. Heat can be transferred by **three mechanisms**: **conduction**, **radiation**, and **convection**. In conduction, the matter in the two regions is stationary and the heat is transferred through direct contact. In radiation, heat is transferred through the emission and absorption of **electromagnetic waves**. Finally, in convection, heat moves via the flow of matter (eg, a gas or a fluid) over large scales.

In this question, the flame in the hot air balloon increases the temperature of the air inside the balloon. As the air's temperature increases, the air becomes **less dense** than the surrounding air outside the balloon. Hence, there is a **buoyant force** that causes the air in the balloon to rise. Therefore, the convection associated with the buoyancy of the hotter, less dense air causes the balloon to move upward.

(Choice B) Cooler air becomes denser and therefore must sink downward instead of rising upward.

(Choice C) The flame does transfer some heat via radiation, but the radiation does not push the air upward.

(Choice D) The air in the balloon is not stationary, so the heat transfer does not occur via conduction.

Educational objective:

Convection occurs due to buoyant forces in fluids. Hotter, less dense air rises, while cooler, denser air sinks.

Time spent:0 sec

Subject:
Physics

QID:404112

System:
Fluids

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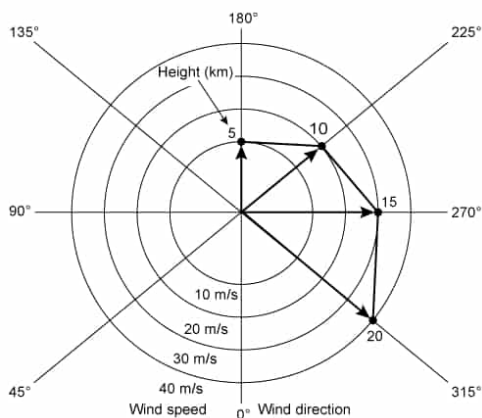
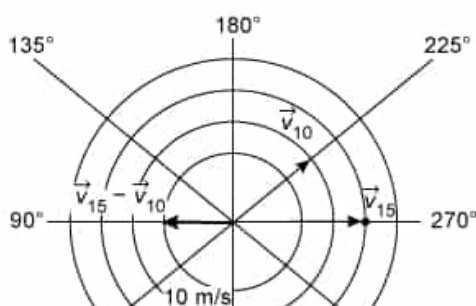


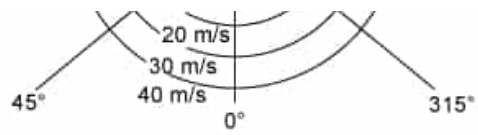
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Which image best illustrates the vector difference $\vec{v}_{15} - \vec{v}_{10}$ between the wind velocity at 15 km and the wind velocity at 10 km in Figure 1?

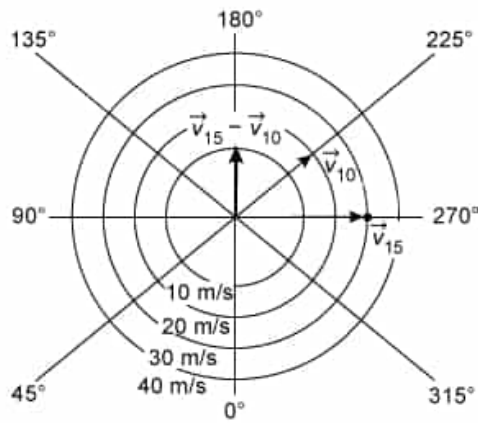
☐ A.





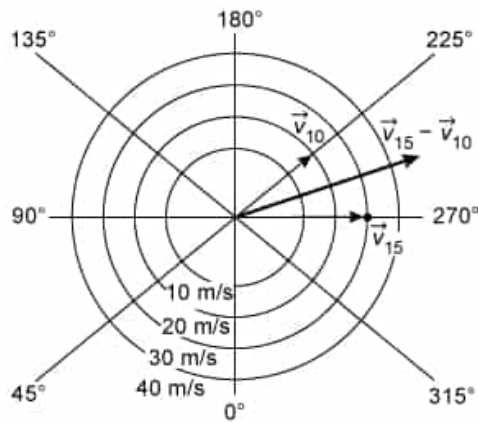
[11%]

☐ B.



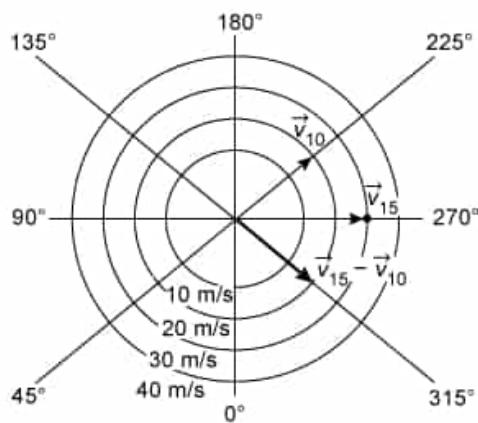
[16%]

☐ C.



[26%]

✓ ☒ D.

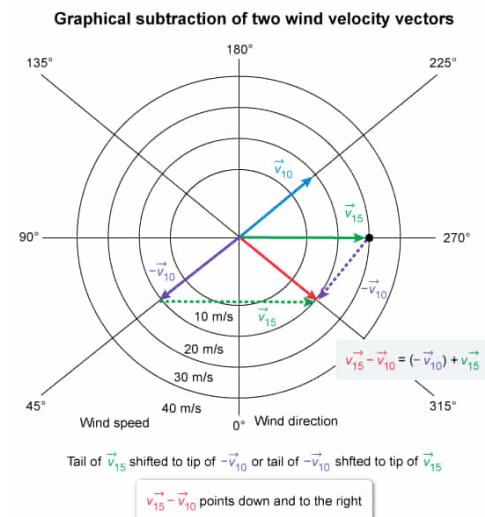


[45%]

Correct

44% answered correctly

Explanation:



Vector quantities (eg, position, velocity, acceleration) can be represented graphically by arrows that show their magnitude and direction. The length of the arrow represents the vector's magnitude, and the direction of the arrow is aligned with the direction of the vector.

The sum of two vectors $\vec{v}_A$ and $\vec{v}_B$ can be determined using the **tip-to-tail method**. In this approach, the tail of the arrow representing $\vec{v}_A$ is shifted such that it touches the tip (ie, head) of the arrow representing $\vec{v}_B$. The sum $\vec{v}_A + \vec{v}_B$ is then represented by the arrow connecting the tail of $\vec{v}_B$ to the tip of the shifted $\vec{v}_A$. The **subtraction** of the two vectors $\vec{v}_A - \vec{v}_B$ can be performed by reversing the direction of $\vec{v}_B$ and then applying the tip-to-tail method: $\vec{v}_A + (-\vec{v}_B)$.

In this question, the wind velocity at 15 km $\vec{v}_{15}$ points to the right with a magnitude of 30 m/s and the wind velocity at 10 km $\vec{v}_{10}$ points up and to the right with a magnitude of 20 m/s. Reversing the direction of $\vec{v}_{10}$ (so that the arrow points downward and to the left) and using the tip-to-tail method yield a velocity difference $\vec{v}_{15} - \vec{v}_{10}$ that points **downward and to the right** (Choice D).

(Choices A and B) These arrows result from incorrectly applying the tip-to-tail method.

(Choice C) This arrow represents the **vector sum** $\vec{v}_{15} + \vec{v}_{10}$, not the difference between the two vectors.

Educational objective:

The difference between two vectors $\vec{v}_A$ and $\vec{v}_B$ can be found by reversing the direction of $\vec{v}_B$, shifting $\vec{v}_A$ so its tail touches the tip of $\vec{v}_B$, and drawing the resultant arrow from the tail of $-\vec{v}_B$ to the tip of $\vec{v}_A$.

Time spent: 0 sec

Subject:

Physics

QID: 404113

System:

Fluids

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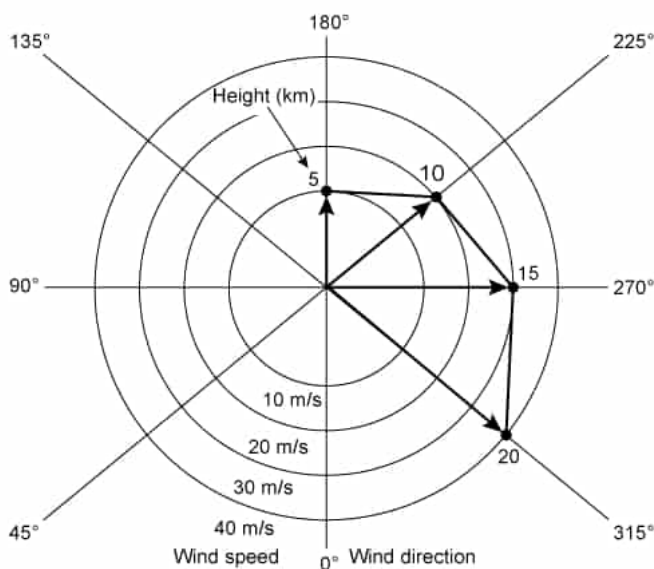


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Hodographs indicate wind shear in the atmosphere, which is an important factor in tornado formation.

When the weather balloon generating the data in Figure 1 reaches an altitude of 20 km, it stops rising and moves horizontally with the wind for 2 hours.

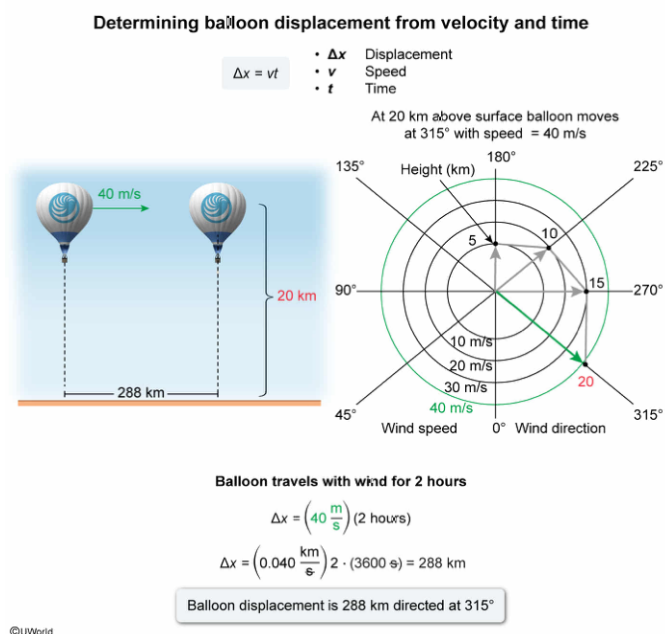
What is the balloon's displacement during this time?

- ☐ A. 4.80 km at 270° [8%]
- ☐ B. 4.80 km at 315° [18%]
- ☐ C. 288 km at 270° [19%]
- ✓ ☒ D. 288 km at 315° [53%]

Correct

54% answered correctly

Explanation:



When an object moves at a constant velocity v over a time t , the object's **displacement** (ie, change in position) Δx equals the product of v and t :

$$\Delta \vec{x} = \vec{v} t$$

In this question, when the weather balloon reaches 20 km it moves horizontally with the wind, which according to Figure 1 has a **speed** of 40 m/s and is directed at 315°. Consequently, the magnitude of the balloon's displacement over an interval of two hours equals:

$$\Delta x = \left(40 \frac{\text{m}}{\text{s}}\right)(2 \text{ hours})$$

Converting hours to seconds yields:

$$(2 \text{ hours}) \left(\frac{60 \text{ minutes}}{1 \text{ hour}} \right) \left(\frac{60 \text{ seconds}}{1 \text{ minute}} \right) = 2(3,600) \text{ s} = 7,200 \text{ s}$$

Hence:

$$\Delta x = \left(40 \frac{\text{m}}{\text{s}}\right)(7,200 \text{ s}) = 288,000 \text{ m}$$

Finally, using $1,000 \text{ m} = 1 \text{ km}$ gives:

$$\Delta x = 288 \text{ km}$$

Therefore, the balloon's displacement is 288 km at a direction of 315° .

(Choices A, B, and C) 4.80 km results from neglecting the extra factor of 60 in the conversion from hours to seconds. 270° results from misreading the hodograph figure.

Educational objective:

For a constant velocity, displacement equals the product of the velocity and time.

Time spent: 0 sec

Subject:

Physics

QID: 404114

System:

Fluids

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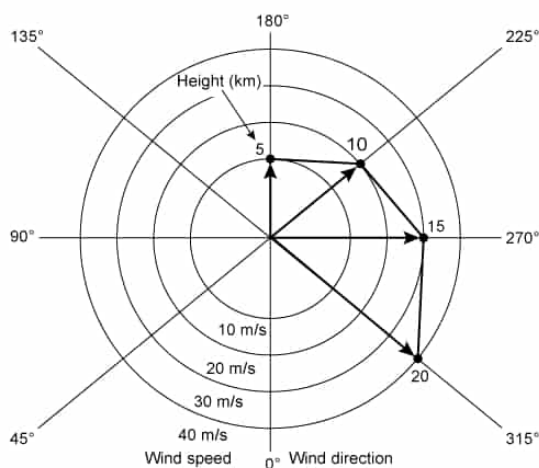


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Which of the following best describes the behavior of air parcels in the atmosphere when the value of the CAPE is negative?

- ☐ A. The parcels remain stationary. [3%]
- ☐ B. The parcels move horizontally. [2%]
- ☐ C. The parcels rise upward. [8%]
- ✔ ☒ D. The parcels sink downward. [85%]

Correct

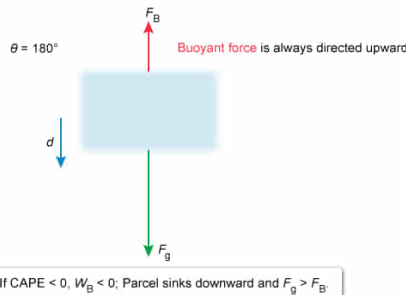
85% answered correctly

Explanation:

Air parcel sinks downward if work done by buoyant force is negative

$$W = Fd \cos \theta$$
$$\text{CAPE} = \frac{W_B}{m}$$

- W Work
- F Force
- d Displacement
- θ Angle between force and displacement vectors
- CAPE Convective available potential energy
- m Mass of air parcel
- W_B Work done by buoyant force



The **work** W done on an object equals the product of the magnitude of the force F exerted on the object, the magnitude of the object's displacement d , and the cosine of the angle θ between the force and displacement vectors:

$$W = Fd \cos \theta$$

Consequently, the value of W is positive if F and d are parallel ($\theta = 0^\circ$) but can be negative if F and d are anti-parallel (ie, oppositely directed, $\theta = 180^\circ$).

In this question, two forces act on the air parcels: the upward directed **buoyant force** F_B due to the displacement of the air surrounding the parcel and the downward directed parcel **weight** F_g . **Newton's second law of motion** implies that an air parcel's behavior is determined by the net force:

$$F_{\text{net}} = F_B - F_g$$

If the buoyant force is greater than the parcel weight F_g , the parcel rises upward in the atmosphere. However, if the buoyant force is smaller than the F_g , the parcel sinks downward.

If the CAPE is negative, the work done by the buoyant force W_B is also negative:

$$\text{CAPE} < 0 \Rightarrow W_B < 0$$

Therefore, the parcels must **sink downward** because the buoyant force is always directed upward.

(Choice A) If the parcels remained stationary, the work done would be zero because the displacement would be zero.

(Choice B) If the parcels moved horizontally, the work done would be zero, because the displacement and force vectors would be perpendicular ($\theta = 90^\circ$).

(Choice C) Upward parcel motion is associated with a positive CAPE because displacement is parallel to the buoyant force.

Educational objective:

The sign of work depends on the alignment of the force and displacement vectors. When work is negative, displacement is opposite to the applied force.

Time spent:0 sec**Subject:**

Physics

QID:404115**System:**

Fluids